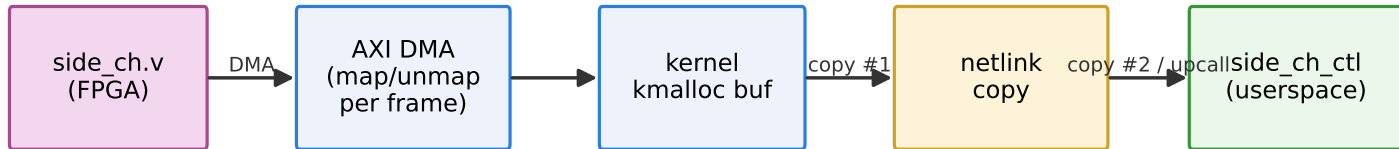
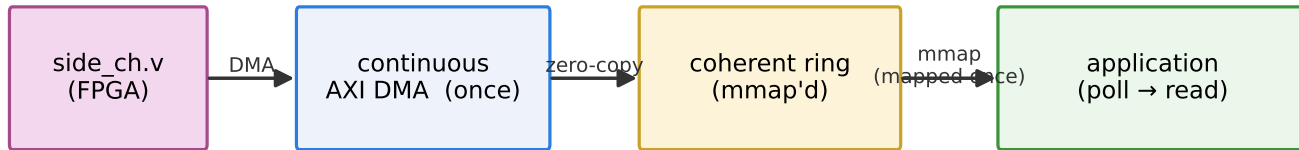


### (a) Stock openwifi CSI path



poll every ~100 ms · CPU copy ×2 · per-frame map/unmap

### (b) Proposed zero-copy CSI path



1× DMA · 0× CPU copy · interrupt-driven